

VELIDANDA KRISHNA SAI

International Institute of Information Technology, Bangalore

☎ +91 70930 79149 ✉ kvelidanda.1177@gmail.com 🔗 [linkedin/krishna-sai-velidanda](https://www.linkedin.com/in/krishna-sai-velidanda) 🐙 [github/melohub-xbit](https://github.com/melohub-xbit)

Education

International Institute of Information Technology, Bangalore

Jul 2023 – Jul 2028

Dual Degree (B.Tech + M.Tech) in Computer Science and Engineering

CGPA: 3.51/4.0 📄

Experience

Research Work – Multimodal Stress Detection

Jan 2026 – May 2026

Samsung Lab, IITB | Supervisor: Dr. Sakshi Arora

Bangalore, India

- Designed a multi-stressor experimental protocol and recorded simultaneous 8-channel EEG (500 Hz) and single-lead ECG from 15 subjects to quantify physiological stress and recovery.
- Engineered a 4,939-window feature extraction pipeline computing EEG band power (delta-gamma), spectral entropy, frontal alpha asymmetry, and ECG-derived HRV metrics (**RMSSD**, **pNN50**, **LF/HF ratio**).
- Trained **SVM** and **CNN-LSTM** classifiers with Leave-One-Subject-Out cross-validation, achieving **95.6%** stress detection accuracy with the CNN-LSTM architecture, demonstrating subject-independent generalization.

Achievements

- First Prize:** ICDEC'24 International Vehicle Detection Challenge – 1st of 2,000 teams.
- National Runner-Up:** TruthTell Hackathon (WAVES Summit 2025) – Top 2 of 5,600+ global submissions. 📄
- 3rd Runner-Up:** Google Gen AI Exchange Hackathon 2025 – 4th among 270,000+ developers. 📄
- 2nd Prize:** MERNify, MERN Stack Hackathon at Synergy'24 – 3,500+ participants.
- Finalist:** Meta x OpenEnv Hackathon, Agentic AI and Reinforcement Learning Hackathon – 2000+ teams.
- Dean's Merit List:** Academic Year 2023–2024. 📄

Projects

Relay Brain 🧠 | FastAPI, Python, Agentic AI, LLMs

May 2026 – June 2026

- Built an AI-assisted knowledge orchestration platform for structured task planning, retrieval, and multi-agent coordination.
- Integrated **LLM-based** reasoning and workflow automation to support contextual decision-making and intelligent agent interactions.
- Developed the system with a modern web interface and backend services for scalable, modular AI-driven execution.

Matrix of Truth 🧠 | Python, FastAPI, Gemini AI SDK, GCP, PyTorch

Sep 2025 – Nov 2025

- Built a **multi-modal misinformation** detection platform with claim extraction, multi-layer LLM fact-checking, and a dynamic knowledge graph for tracking entity-level misinformation patterns.
- Integrated **deepfake detection** for video, image, and audio, reverse image search via Cloud Vision API, and real-time scam alert monitoring using Gemini search grounding.
- Deployed on **Google Cloud Run** with a React dashboard and Chrome extension.

Dialecto 🧠 | FastAPI, MongoDB, Agentic AI, GenAI

Nov 2024 – Dec 2024

- Built a gamified language-learning platform using **FastAPI** and **MongoDB**, integrating AI-driven storytelling and adaptive feedback loops.
- Designed interactive mini-games and conversational exercises powered by **GenAI** to personalize grammar, vocabulary, and comprehension practice.
- Implemented an **agentic tutoring workflow** for real-time response generation and engagement tracking, improving learner interaction quality and retention.

High-Frequency Trading Simulation 🧠 | C++, Java

Oct 2024 – Dec 2024

- Engineered an order matching engine in C++ with separate buy and sell **AVL trees**; exposed to Java via JNI for cross-language integration.
- Implemented **market-making** and **cross-exchange arbitrage** strategies, run concurrently via multithreading.
- Added a **risk management** module with stop-loss halts, inventory tracking to prevent overselling, and Sharpe ratio computation for evaluating risk-adjusted returns per strategy.

Technical Skills

Languages: Python, C++, C, Java, SQL, JavaScript

Frameworks/Databases: PyTorch, FastAPI, MongoDB, MySQL, React.js, Node.js

Tools & Platforms: Docker, Google Cloud Platform, Git, Linux

Relevant Coursework

Data Structures & Algorithms, OOP (C++ & Java), Operating Systems, Computer Networks, Database Systems, Natural Language Processing, Linear Algebra, Software Engineering, Machine Learning